Galois Representations from Torsion of Abelian Surfaces

Aidan Hennessey and Mattie Ji

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These are notes taken live by Mattie Ji at Aidan Hennessey's talk *Galois Representations from Torsion of Abelian Surfaces*, given at Brown's Undergraduate Math Colloqium in October 2024, edited and posted by Aidan with permission from Mattie. A huge, huge thank-you to Mattie for writing up the first draft. This talk is based on work this summer joint with Andy Zhu and Mathilde Kermorgant, supervised by Isabel Vogt and Sachi Hashimoto.

This talk is about abelian varieties!

Definition 0.1. An **abelian variety** is a compact (i.e. projective) algebraic curve, surface, or higher dimensional analogue with a "nice" abelian group structure.

1 Elliptic Curves

We will begin the talk by talking about elliptic curves. The importance of elliptic curves comes from the following theorem.

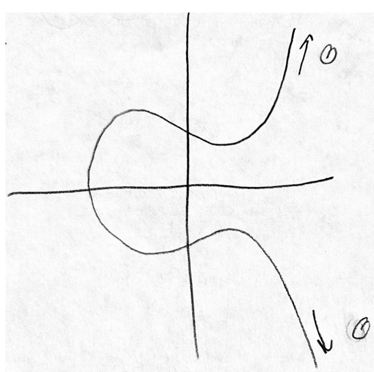
Theorem 1.1. Every 1-dimensional abelian variety is (isomorphic to) an elliptic curve.

Definition 1.2. An **elliptic curve** is (outside of characteristics 2 and 3) the set of solutions (x, y) to an equation of the form

$$E : y^2 = x^3 + Ax + B$$

where $4A^3 + 27B^2 \neq 0$ (this condition ensures that the left-hand side lacks repeated roots), together with a "point at ∞", which we shall denote $\mathcal{O}$. This point is added to compactify the curve.

The picture to keep in mind of an elliptic curve is as follows



Here, the point at ∞, $\mathcal{O}$, should be thought of as if it lies on every vertical line.

We mentioned earlier that elliptic curves are abelian varieties, so there should be a group law on their points. We may define such a group as follows.

Definition 1.3. To an elliptic curve E, we associate the group $\mathrm{Jac}(E)$, which is a quotient of the free abelian group on the points of E by the following relations.

1. Pick an identity. Any point will work, but it is standard to pick the point at ∞, $\mathcal{O}$.
2. The sum of three colinear points should be 0.

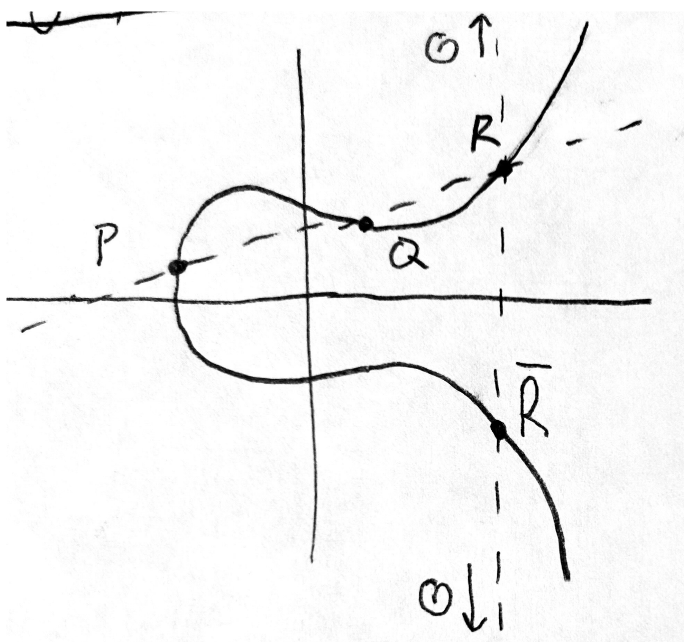
The points of E are in bijection with the elements of $\mathrm{Jac}(E)$. Concretely, we add points of E as follows. For any $P = (x_1, y_1), Q = (x_2, y_2) \in E$, the line through P, Q intersects the elliptic curve E at a unique third point R, and we write

$$P + Q + R = 0.$$

Let $\overline{R}$ denote the unique third intersection of E with the vertical line going through R (the first two being R and $\mathcal{O}$). Since $R + \overline{R} + \mathcal{O} = \mathcal{O}$, R and $\overline{R}$ are inverses. We thus may write

$$P + Q = \overline{R}.$$

There are some edge cases to this. If $P = Q$, then we take the tangent line to the curve at P to be the 'line through P and Q'. Also, if two distinct points lie on the same vertical line, their sum will be $\mathcal{O}$.



Notice that if $P, Q \in E$ have coordinates in K for some field K, and K also contains the coefficients A and B defining E, then the cubic giving the x-coordinates of the points of the intersection $E \cap \ell$ has coefficients in K. Thus, the sum of the roots of the cubic lies in K, as do two of the roots, so the third must as well. The x-coordinate of $P + Q$ is thus in K, and it follows that the y-coordinate of $P + Q$ is as well (modulo some discussions about infinity). Thus, for any field containing A and B, we get a group.

Definition 1.4. Let K be a field. Let $E : y^2 = f(x)$ for $f(x) \in K[x]$ an appropriate cubic. Then, define $E(K)$ to be the **group** of K-points, i.e. points with coordinates in K.

2 Jacobians of Hyperelliptic curves

1-dimensional abelian varieties are elliptic curves. To get higher-dimensional abelian varieties, we may look at *hyper*elliptic curves.

Theorem 2.1. Every (principally polarzed) abelian surface is, up to isomorphism, either

1. a product of two elliptic curves, or
2. the Jacobian of a (genus 2) hyperelliptic curve.

Remark 2.2. This Jacobian variety construction is much more general and can be used to generate general abelian varieties, whose dimension depends on the genus of the hyper-elliptic curve.

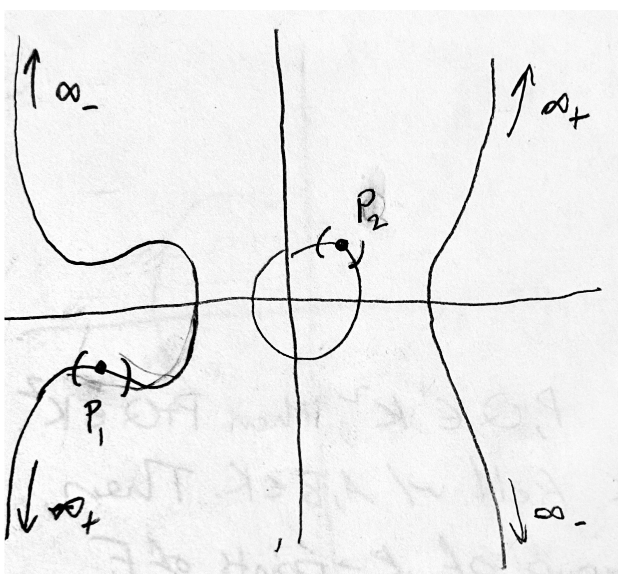
What is the Jacobian of a hyperelliptic curve? We will not be super rigorous, but we give enough of an idea such that our most assiduous audience members will be able to iron out the details on their own.

Definition 2.3. A (genus 2) hyperelliptic curve is the set of solutions (x, y) to the equation

$$C : y^2 = f(x), \quad \deg(f) = 6, \ \mathrm{Disc}(f) \neq 0$$

together with, in contrast to the elliptic curve case, *two* points at ∞, which we call ∞_+ and ∞_-

Below is an example, with infinities labeled.



Definition 2.4. The **Jacobian** J of a (genus 2) hyper-elliptic curve is, *roughly* speaking, the space of **unodered pairs of points** on C. Since, locally, this space looks like the product of small intervals around each point, this is a surface. The Jacobian is equipped with a group law as follows:

1. Every vertical pair (pair of points lying on the same vertical line) is 0.
2. Any triplet of pairs, all 6 of whose points lie on the graph of a single "cubic", adds to 0.

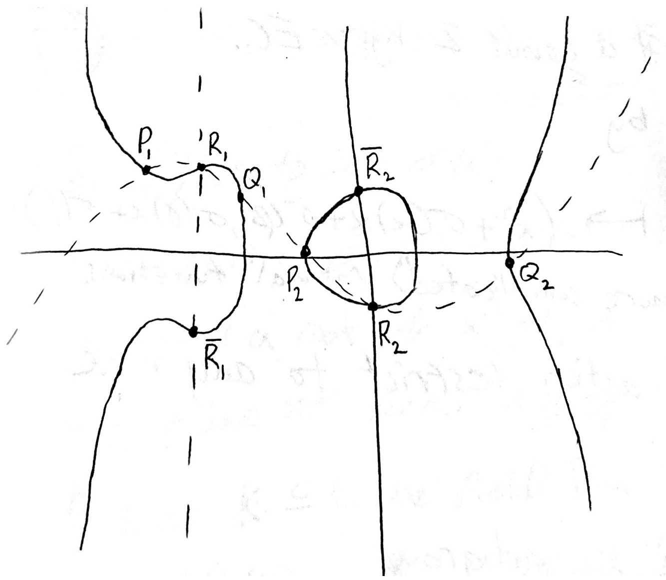
Just as there is a unique line interpolating any two distinct points, there is a unique cubic interpolating any four points (to deal with points whose x-coordinates are the same, we broaden the notion of "graph of a cubic" to include triplets of vertical lines). Consider two pairs P, Q. For algebraic reasons, their interpolating cubic will intersect the hyperelliptic curve at 2 other points (possibly at ∞). Call this third pair R.

As in the elliptic curve case, we will have that

$$P + Q + R = 0, \quad R + \overline{R} = 0,$$

and thus conclude that

$$P + Q = \overline{R}.$$



The next thing we did for elliptic curves, after defining the group law, was to construct groups associated to each field over which the curve is defined.

Definition 2.5 (Attempted Definition)**.** Let K be a field for which $f(x) \in K[x]$, then the Jacobian $J(K)$ is the set of point pairs for which points have K-rational coordinates.

This may seem like a reasonable definition, but it is actually somewhat problematic. Consider the process of computing the sum $P + Q$. To find the x-coordinates of the sum, we compute the interpolating cubic $g(x)$, square it, and set it equal to the defining polynomial $f(x)$. We already know 4 roots of the resulting polynomial (the x-coordinates of the points in the pairs P and Q), so we factor those out, leaving something like $f(x) - g^2(x) = (x - x_1)(x - x_2)(x - x_3)(x - x_4)(x^2 - \alpha x + \beta)$. Solving the quadratic may require passing to an extension; the x-coordinates of the points of R need not live in the same field as those of P and Q. Thus, our attempted definition does not yield a group.
Fortunately, the solution to this is given by what are called "**Mumford coordinates**".

Definition 2.6. Let $P = \{(x_1, y_1), (x_2, y_2)\}$, the **Mumford coordinates of** P are a pair

$$(x^2 + \alpha x + \beta, \gamma x + \delta)$$

such that

1. $x^2 + \alpha x + \beta = (x - x_1)(x - x_2)$
2. $y_1 = \gamma x_1 + \delta$ and $y_2 = \gamma x_2 + \delta$

Mumford coordinates are great because they bypass the need to solve the above quadratic equation. We leave it to the interested viewer to work out how to compute point addition in terms of Mumford coordinates, and we implore such viewers to notice how one never needs to pass to an extension. This motivates an improved definition for $J(K)$.

Definition 2.7. Let K be a field such that $f(x) \in K[x]$. Then, $J(K)$ is the **group** of point-pairs whose Mumford coordinate coefficients $(\alpha, \beta, \gamma, \delta)$ are contained in the field K.

3 Galois Representations

This talk was marketed as accessible to students with only a 1-semester algebra background, so in honor of that we now take a moment to give a brief crash course on Galois theory, starting with the following question.

Question 3.1. What are the field automorphisms $\mathbb{C} \to \mathbb{C}$ fixing $\mathbb{R}$?

Answer: The identity map and the complex conjugation. In fact, those are the only two.

To see why, consider an automorphism σ of $\mathbb{C}$ fixing $\mathbb{R}$, then we have that

$$\sigma(a + ib) = \sigma(a) + \sigma(ib) = a + b\sigma(i).$$

Thus σ is completely determine by its action on i. On the other hand since $i^2 = -1$, we have that

$$-1 = \sigma(-1) = \sigma(i^2) = (\sigma(i))^2.$$

Thus, we see that either $\sigma(i) = i$ (so it is the identity) or $\sigma(i) = -i$ (so it is conjugation).

More generally, we have the following construction.

Definition 3.2. Let $K \subseteq L$ be fields, the Galois group $\mathrm{Gal}(L/K)$ is the set of all field automorphisms of L fixing K (with the group law being composition).

Fact: The Galois group of L/K permutes the L-roots of $p(x) \in K[x]$.

The fields we want to be considering are as follows.

Definition 3.3. The set of roots to polynomials $f(x) \in \mathbb{Q}[x]$ form a field called the *algebraic closure of* $\mathbb{Q}$, denoted $\overline{\mathbb{Q}}$.

Proposition 3.4. The Galois group $\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$ acts on $E(\overline{\mathbb{Q}})$ for an elliptic curve E. Furthermore, this action respects the group structure of $E(\overline{\mathbb{Q}})$. The action is coordinate-wise, i.e.

$$\sigma \cdot (x, y) = (\sigma(x), \sigma(y)).$$

Proof Idea. Let $E : y^2 = x^3 + Ax + B$ be an elliptic curve, with $(x_1, y_1), (x_2, y_2) \in E$. Then, one may compute explicitly that the sum (x_3, y_3) is given by rational functions, specifically,

$$x_3 = \left(\frac{y_2 - y_1}{x_2 - x_1}\right)^2 - (x_1 + x_2) \quad \text{and} \quad y_3 = \left(\frac{y_2 - y_1}{x_2 - x_1}\right)(x_1 - x_3) - y_1.$$

From here, one can successively apply the automorphism rules to conclude that $\sigma(x_1, y_1) + \sigma(x_2, y_2) = \sigma(x_3, y_3)$. ∎

Proposition 3.5. Let J be the Jacobian of a genus 2 hyper-elliptic curve. Then, similarly, the Galois group $\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$ acts on $J(\overline{\mathbb{Q}})$ by

$$\sigma(x^2 + \alpha x + \beta, \gamma x + \delta) = (x^2 + \sigma(\alpha)x + \sigma(\beta), \sigma(\gamma)x + \sigma(\delta)).$$

Like in the elliptic curve case, addition is given by rational maps, and rational maps commute with Galois automorphisms, so this action also respects the group law on the Jacobian.

The Jacobian over $\overline{\mathbb{Q}}$ is a large, unwieldy beast, so for purpose of study, we like to focus on special, smaller subgroups of $J(\overline{\mathbb{Q}})$. Here is one possible choice, which is not just countable, and not just finitely generated, but in fact finite!

Definition 3.6. The n-torsion subgroup $J(\overline{\mathbb{Q}})[n]$ is the subgroup

$$\{P \in J(\overline{\mathbb{Q}}) \mid nP = 0\}.$$

This construction is applicable to any abelian group.

Fact: $J(\overline{\mathbb{Q}})[n]$ is isomorphic to $(\mathbb{Z}/n\mathbb{Z})^4$.

Not only do we have this finite subgroup, we also claim that there is an action of $\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$ on $J(\overline{\mathbb{Q}})[n]$ by restriction. The reason why is because the map $P \mapsto nP$ is a (very complicated) rational map.

Recall: The data of a group action G on some structure X is equivalent to a group homomorphism $G \to \mathrm{Aut}(X)$.

For $n = \ell$ a prime, we have that

$$J(\overline{\mathbb{Q}})[\ell] \cong (\mathbb{Z}/\ell\mathbb{Z})^4 \cong \mathbb{F}_\ell^4.$$

In this case, the action of the Galois group $\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$ is equivalent to a map

$$\rho : \mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q}) \to \mathrm{GL}_4(\mathbb{F}_\ell).$$

Definition 3.7. The mod-ℓ **Galois representation** from torsion of an abelian surface is the map

$$\rho_{J,\ell} : \mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q}) \to \mathrm{GL}_4(\mathbb{F}_\ell)$$

given by the correspondence as above.

Remark 3.8. Our specific project this summer was to compute the image of this map for the case $\ell = 5$.

Remark 3.9. The mod-ℓ Galois representation is useful to compute because it encodes information about what kinds of maps can go into or come out of a Jacobian. For example, whether there is a map out of J whose kernel is $\mathbb{Z}/\ell\mathbb{Z}$ is encoded in the mod-ℓ Galois representation.